



LONGEVITY & ANTI-AGING

Epithalon

Epithalon – 50 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	L-Alanyl-L-glutamyl-L-aspartyl-glycine
SYNONYMS / ALIASES	Epithalon; Epitalon; AEDG peptide; Khavinson tetrapeptide
CAS NUMBER	307297-39-8
MOLECULAR FORMULA	C14H22N4O9
MOLECULAR WEIGHT	390.35 g/mol
AMINO ACID SEQUENCE	AEDG (4 residues; H-Ala-Glu-Asp-Gly-OH)
SOURCE	Synthetic (solid-phase peptide synthesis); sequence designed as a short structural analog of bovine pineal-derived oligopeptide preparation (Epithalamin)

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	>=99% by HPLC (confirm against batch COA)
CONTAINER	50 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	-20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C, stable approximately 28 days in bacteriostatic water
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only)
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months at -20 C
STABILITY NOTES	Small linear tetrapeptide; generally stable. Avoid repeated freeze-thaw cycles.

STRUCTURAL NOTES

Epithalon is a synthetic tetrapeptide (4 amino acids) of sequence Ala-Glu-Asp-Gly. It was developed as a defined short-peptide analog of Epithalamin, a heterogeneous polypeptide preparation originally isolated from the bovine pineal gland. Epithalon belongs to the "Khavinson peptide" family of short regulatory peptides (typically tri- or tetrapeptides) characterized in Russian peptide research literature.

SELECTED PEER-REVIEWED REFERENCES

1. Khavinson V.K. - *Peptides and Ageing* - Neuroendocrinology Letters, 2002 - PMID: 12374906
2. Khavinson V.K., Bondarev I.E., Butyugov A.A. - *Epithalon peptide induces telomerase activity and telomere elongation in human somatic cells* - Bulletin of Experimental Biology and Medicine, 2003 - DOI: 10.1023/B:BEBM.0000007649.83476.b8
3. Anisimov V.N. et al. - *Effect of Epitalon on biomarkers of aging, life span and spontaneous tumor incidence in female Swiss-derived SHR mice* - Biogerontology, 2003 - DOI: 10.1023/A:1024676925150

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

COMPLIANCE DISCLAIMER

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Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.